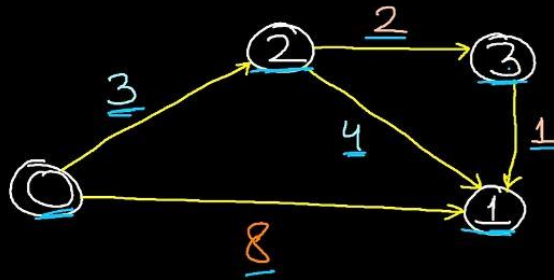
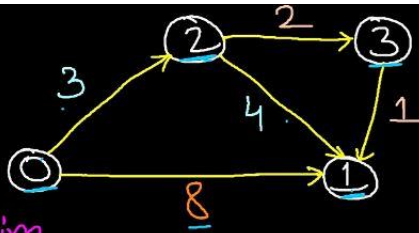


Floyd Warshall All Pair Shortest Path Algorithm



$$\cancel{d[0][1] = 8} \neq \textcircled{6}$$

NOTE: We include each vertex one by one in calculation of shortest path.
As soon as we find a shorter path, we update



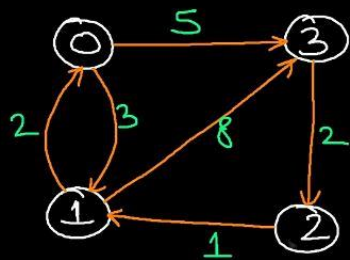
$$5+1 = \underline{\underline{6}} \quad \textcircled{k^{\text{th}}}$$

Including

$$\left\{ \begin{array}{l} \underline{2} \quad d[0][1] = \min(d[0][1], d[0][2] + d[2][1]) \\ \quad \quad = \min(8, 3+4) = \textcircled{7} \\ \underline{3} \quad d[0][1] = \min(d[0][1], d[0][3] + d[3][1]) \\ \quad \quad = \min(7, 5+1) = \textcircled{6} \end{array} \right.$$

Via k:-

$$\underline{d[i][j]} = \min(\underline{d[i][j]}, \underline{d[i][k]} + \underline{d[k][j]}) \quad \neq$$



D^0

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	8
2	∞	1	0	∞
3	∞	∞	2	0

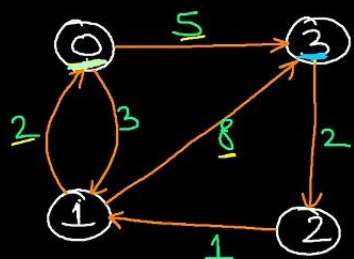
$D^0 \Rightarrow$

	0	1	2	3
0				
1				
2				
3				

$\Downarrow D^1$

	0	1	2	3
0				
1				
2				
3				

NOTE: Distances don't get modified for k^{th} adjacent vertices



D^0

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	8
2	∞	1	0	∞
3	∞	∞	2	0

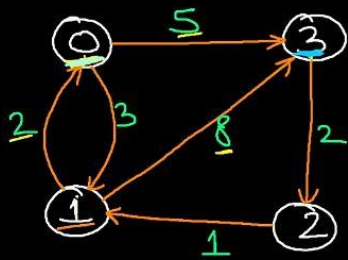
$D^0 \Rightarrow$

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	7
2	∞	1	0	∞
3	∞	∞	2	0

$\Downarrow D^1$

	0	1	2	3
0				
1				
2				
3				

$$\begin{aligned}
 d[1][2] &= \min(d[1][2], d[1][0] + d[0][2]) \\
 &= \min(\infty, 2 + \infty) \\
 &= \infty \\
 \min(8, 2 + 5) &= 7 \quad \min(\infty, \infty + \dots) = \infty \\
 &\quad \min(2, \infty + \dots)
 \end{aligned}$$



D

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	8
2	∞	1	0	∞
3	∞	∞	2	0

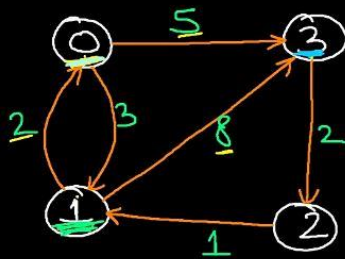
$\Rightarrow D^0$

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	7
2	∞	1	0	∞
3	∞	∞	2	0

$\downarrow D^1$

	0	1	2	3
0	0	3		
1	2	0	∞	7
2		1	0	
3		∞		0

NOTE: Distances don't get modified for k^{th} adjacent vertices.



D

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	8
2	∞	1	0	∞
3	∞	∞	2	0

$\Rightarrow D^0$

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	7
2	∞	1	0	∞
3	∞	∞	2	0

$\downarrow D^1$

	0	1	2	3
0	0	3	∞	5
1	2	0	∞	7
2	3	1	0	8
3	∞	∞	2	0

$$\begin{aligned} \min(\infty, 3 + \infty) &= \infty \\ \min(5, 3 + 7) &= 5 \\ \min(\infty, 1 + 2) &= 3 \\ \min(\infty, 1 + 7) &= 8 \end{aligned}$$

$$\begin{aligned} \min(\infty, \infty + \dots) &= \infty \\ \min(2, \infty + \dots) &= 2 \end{aligned}$$

D^2		0	1	2	3
0	0	3	∞	5	
1	2	0	∞	7	
2	3	1	0	8	
3	5	3	2	0	

 $\Rightarrow$

D^3		0	1	2	3
0	0	3	7	5	
1	2	0	9	7	
2	3	1	0	8	
3	5	3	2	0	

Shortest Path from i to j (which may or may not include k)

$$\underline{d[i][j]} = \min(\underline{d[i][j]}, \underline{d[i][k]} + \underline{d[k][j]})$$

All Pairs Shortest Path

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & \\ 2 & & & 0 \end{bmatrix}$$

$$A^0[2,3] \quad A^0[2,1] + A^0[1,3]$$

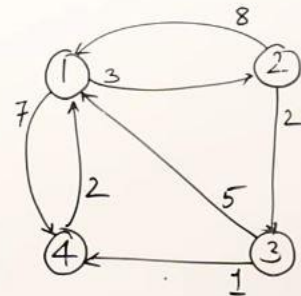
$$2 < 8 + \infty$$

$$A^0[2,4] \quad A^0[2,1] + A^0[1,4]$$

$$\infty > 8 + 7$$

$$A^0[3,2] \quad A^0[3,1] + A^0[1,2]$$

$$\infty > 5 + 3$$



$$A^0 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & \infty \\ 5 & 8 & 0 & 1 \\ 2 & \infty & \infty & 0 \end{bmatrix}$$

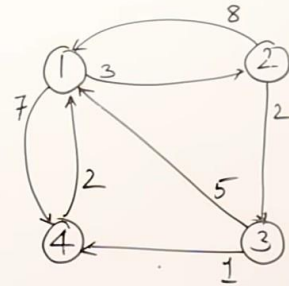
All Pairs Shortest Path

$$A^1 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & \infty & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & 5 & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & 5 & 6 \\ 7 & 0 & 2 & 3 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & 5 & 6 \\ 5 & 0 & 2 & 3 \\ 3 & 6 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$



$$A^0 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & \infty \\ 5 & 8 & 0 & 1 \\ 2 & \infty & \infty & 0 \end{bmatrix}$$

$$A^k[i, j] = \min \left\{ \underbrace{A^{k-1}[i, j]}_{\text{direct path}}, \underbrace{A^{k-1}[i, k] + A^{k-1}[k, j]}_{\text{path via } k} \right\}$$